

Attention Is All You Need

Ashish Vaswani, Noam Shazeer, Niki Parmar

Abstract

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms.

1. Introduction

Recurrent neural networks, long short-term memory and gated recurrent neural networks have been firmly established as state of the art approaches in sequence modeling and transduction problems.

2. Self-Attention Mechanism

An attention function can be described as mapping a query and a set of key-value pairs to an output. The output is computed as a weighted sum.

3. Results

On the WMT 2014 English-to-German translation task, the big transformer model outperforms the best previously reported models by more than 2.0 BLEU.